

Stage 3 — Real artifacts across modalities

Where the Shortcut Sits: progress report 3

In one sentence. With real, unedited artifacts — hair in dermoscopy, calipers in thyroid and ovarian ultrasound, debris in capsule endoscopy — ROI masking helps far more when the artifact lies outside the ROI than inside it: the crossover is positive in 9 of 10 cohort–encoder pairs and in all four pre-specified location tests after Holm correction; the dermatology encoder DermLIP and, with the radiograph encoder RAD-DINO, real chest-radiograph devices are the exceptions, and the hair shortcut turns out to be carried largely by correlates of hair rather than by its pixels.

Notation

$Y \in \{0, 1\}$	diagnosis: 1 = melanoma (dermoscopy), malignant nodule (thyroid), tumour with a solid component (ovary), erosion (capsule), pneumothorax or the finding of interest (chest radiograph).
$A \in \{0, 1\}$	artifact present: a real artifact: hair (dermoscopy), sonographer caliper or measurement mark (thyroid, ovary), debris/bubbles/bile (capsule endoscopy), device or drain (chest radiograph).
ROI	region of interest: lesion, nodule, tumour, union of expert lesion boxes, lungs; drawn in green ; artifacts in red .
r	overlap $ \text{artifact} \cap \text{ROI} / \text{artifact} $. Trap A: $r \geq 0.5$ (artifact in the ROI); Trap B: $r < 0.1$ (outside).
Environments	training and correlated test $P(A=1 Y=1) = 0.9$, $P(A=1 Y=0) = 0.1$; reversed test 0.1/0.9; clean test: artifact independent of Y (real artifacts: half of each class) or absent (synthetic artifacts).
ERM, mask	linear head on the frozen features of the original image / of the ROI-masked image.
Crossover	$[\text{mask} - \text{ERM}]_{\text{Trap B}} - [\text{mask} - \text{ERM}]_{\text{Trap A}}$ of reversed-test AUROC (positive: masking is worth more when the artifact lies outside the ROI).
CI	95% crossed seed×image bootstrap interval (Stage 1); every interval in this report uses it unless marked “not re-estimated”.
$ \Delta p $	same-head counterfactual sensitivity: change of the predicted probability when the artifact is inserted into the same image.

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1 Where this stage starts

Stage 1 showed with real hair (E13) that masking harms when hair lies on the lesion and helps when it lies beside it. Stage 2 established the location law with synthetic artifacts whose position was controlled, in five modalities, and identified its mechanism (retention with amplified reliance). Synthetic artifacts answer *whether* location can matter; they cannot show that it matters for the artifacts clinicians actually produce, whose location is set by the workflow and read from masks. This stage builds real-artifact traps in three more cohorts, validates the automatic artifact masks that define “inside” and “outside”, replicates across encoder families, asks what actually carries the hair shortcut, and examines chest radiography, where the law might not hold.

2 Questions

- Q3a** With real artifacts and the same protocol, is ROI masking less useful — or harmful — when the artifact lies inside the ROI than when it lies outside it (a positive crossover)?
- Q3b** Does this hold across modalities, artifacts and encoder families?
- Q3c** Are the artifact masks that define inside and outside good enough to support the answer?
- Q3d** What carries a real shortcut: the artifact’s pixels, or things that go with the artifact?

Design: two traps sharing one artifact-free group A real artifact’s location is fixed by the image, so it cannot be moved. Each cohort is split into images whose artifact lies mostly inside the ROI ($r \geq 0.5$, Trap A) and images whose artifact lies outside it ($r < 0.1$, Trap B); images with $0.1 \leq r < 0.5$ belong to neither trap. Each trap is completed by the *same* artifact-free images and receives the environments of Stage 1. This is the E13 design of Stage 1, which replaced the pilot’s overlap-contrast design (E12) because in E12 location itself was the label proxy.

Why these cohorts Each has (i) an artifact that clinicians produce and that is associated with the diagnosis through the workflow (calipers are placed on nodules that are measured; debris is more frequent on some lesion types), (ii) public ROI masks or boxes, (iii) a way to locate the artifact without manual annotation, and (iv) enough images for an artifact-free group. Table 13 lists the cohorts that failed one of these conditions.

3 Methods

3.1 Cohorts

- **Dermoscopy** — ISIC 2019 (Combalia et al., 2019; Tschandl et al., 2018) as in Stage 1: published semi-automatic hair masks (Wegley et al., 2026); lesion masks from HAM10000 or the U-Net trained on ISIC 2018 Task 1 (Codella et al., 2019); leakage groups lesion identifier $\cup$ perceptual hash.
- **Thyroid ultrasound** — TN3K images (Gong et al., 2021) with TNCD benign/malignant labels (3,493 images, 1,210 malignant) and manual nodule masks. Calipers located by a rule-based detector (dotted measurement lines = ≥ 7 evenly spaced bright dots; “+” calipers by template matching), tuned on 36 images and frozen before its final audit. $A = 0$: no detected caliper pixel; $A = 1$: ≥ 15 detected pixels. Groups: pHash ≤ 8 (TN3K publishes no patient identifiers).
- **Ovarian ultrasound** — MMOTU OTU_2d (Zhao et al., 2022) (CC BY 4.0): benign cystic lesions (chocolate cyst, serous cystadenoma, simple cyst) versus lesions with a solid component (teratoma, theca cell tumour, mucinous cystadenoma, high-grade serous carcinoma); normal ovaries excluded (1,202 images). The frozen thyroid detector, unchanged. Groups: pHash ≤ 8 .

- **Capsule endoscopy** — SEE-AI (Yokote et al., 2023) (CC BY 4.0), single-class frames: erosion versus polyp-like lesions; ROI = union of the expert lesion boxes. Contamination (debris, bubbles, bile) masks from the published expert masks where available (22 frames) and otherwise from a linear probe on DINOv2 patch tokens trained on 2,160 expert-masked frames. $A = 0$: contamination $< 3\%$ of the frame; $A = 1$: $\geq 10\%$. Groups: blocks of 100 consecutive frame numbers $\cup \text{pHash} \leq 2$ (no video identifiers are published).
- **Chest radiography** — NIH ChestX-ray14 (Wang et al., 2017); lungs from a PSPNet (Cohen et al., 2022). (i) *Device traps*: images linked to RANZCR-CLiP device annotations; Trap A = central venous catheter inside the lungs, Trap B = endotracheal tube outside them; four findings with ≥ 40 reversed-test positives; restricted to AP films (intubated patients are imaged supine, so view could not be matched). (ii) *Chest drains*: 29,687 pneumothorax-labelled images; drains from NEATX human labels and a drain detector (precision 0.95); drains lie inside the lungs, so this is an in-ROI trap only. Patient-grouped folds.

Protocol Frozen encoders: DINOv2 ViT-B/14 @518 for every cohort; MedSigLIP-448 (a medical vision–language model) for thyroid, ovary and capsule; ConvNeXt-Base @384 (a CNN) for thyroid and capsule; DermLIP for dermoscopy; RAD-DINO for chest radiographs. Logistic heads chosen on clean validation data; 5 seeds (42, 123, 456, 789, 2026) $\times$ 5 grouped folds, fold predictions pooled per seed. Within each label the artifact-bearing images are subsampled to the source mix of the artifact-free group (source matching; vacuous for single-source cohorts). Gates fixed before fitting: every matched cell ≥ 25 images and ≥ 40 reversed-test positives. Arms as in Stage 1.

Statistics Crossed seed $\times$ image bootstrap, 10,000 replicates; the crossover’s interval resamples the same images in both traps. The four location tests (P1, one per cohort, DINOv2) are Holm-corrected; everything else is reported with unadjusted 95% intervals.

4 Experiments

Table 1: Experiments of this stage with registration status, the registration document (docs/PREREGISTRATION_<name>.md), its first commit and the support criterion stated in advance. Commit times are set by the committing machine and are not independent proof of order.

Experiment	Status	Registration	First commit (local time)	Support criterion
Dermoscopy hair traps (E13, DINOv2)	supported registered	ISIC2019_TRAPS	76e25d4, 13:14	crossover CI > 0
Dermoscopy hair, DermLIP	supported registered	ISIC2019_TRAPS	76e25d4, 13:14	crossover CI > 0
Thyroid caliper traps	supported registered	THYROID_TRAPS	ea57661, 22:46	crossover CI > 0
Ovarian caliper traps	supported registered	OVARY_TRAPS	2517600, 07:29	crossover CI > 0
Capsule debris traps	supported registered	CAPSULE_TRAPS	fcfb8d9, 00:40	crossover CI > 0
Chest-radiograph device traps	supported registered	CXR_DEVICE_TRAPS	4d1a924, 18:07	crossover CI > 0
Location tests P1 (four cohorts, Holm)	partly supported each test registered; Holm grouping post hoc	STATISTICAL_PLAN	4e9d44c, 09:37	Holm $p < 0.05$, positive estimate
Chest drains (NIH)	supported registered	CXR_DRAIN	3d0cffc, 03:29	mask — ERM reported with CI, no direction
Crossed-bootstrap intervals (A1)	supported registered	FINAL (A1)	c45d482, 00:35	a claim is stated only if its crossed CI excludes 0

5 Results

5.1 Cohorts and artifact masks

Table 2: Real-artifact cohorts and pool sizes (images, positive/negative diagnosis) before the environments are sampled. Trap A: artifact mostly inside the ROI ($r \geq 0.5$); Trap B: outside ($r < 0.1$); both share the artifact-free group.

Cohort	Task	Artifact source)	(mask	ROI	Artifact-free pos/neg	Trap A pos/neg	Trap B pos/neg
ISIC 2019 dermoscopy (HAM10000, BCN20000, MSK)	melanoma vs other	hair (published masks)		lesion: HAM manual / U-Net	157/706	942/2655	969/4412
TN3K ultrasound, TNCD labels	malignant vs benign nodule	calipers (rule detector)		nodule (manual)	749/1052	292/858	99/227
MMOTU 2D ultrasound	solid-component vs cystic tumour	calipers (rule detector)		tumour (manual)	178/170	276/341	103/91
SEE-AI endoscopy	capsule erosion vs polyp-like lesion	debris (patch probe)		lesion box (expert)	343/164	412/404	1009/231

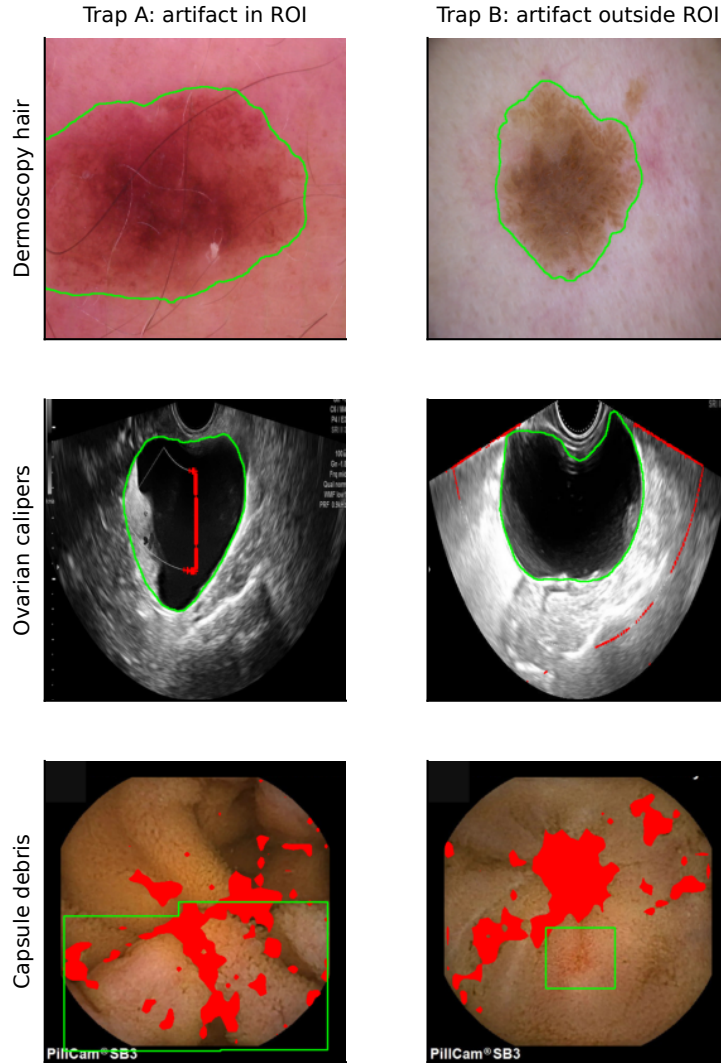


Figure 1: Real artifacts inside (Trap A) and outside (Trap B) the ROI. **Green**: ROI; **red**: automatic artifact mask. Dermoscopy is shown with the lesion outline only (the published hair masks carry no redistribution licence). Thyroid images are not shown (data licence not stated). Automatic overlap of the shown images: dermoscopy 0.534 and 0.000; ovary 1.000 and 0.000; capsule 0.740 and 0.005. ISIC CC BY-NC 4.0; MMOTU, SEE-AI CC BY 4.0.

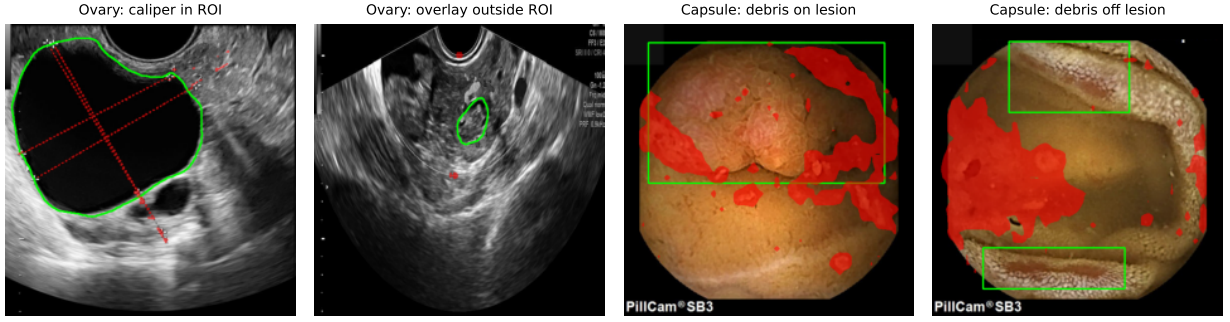


Figure 2: Further real examples with the automatic masks: ovarian calipers crossing the tumour and an overlay outside it; capsule debris on and off the lesion boxes (MMOTU, SEE-AI; CC BY 4.0).

Mask validity Table 3. Lesion masks are accurate (U-Net held-out Dice 0.886; against HAM manual masks 0.895). The capsule probe reaches pixel accuracy 0.922 and IoU 0.625 against held-out expert masks. The thyroid detector’s detections lie on a real caliper in 38 of 40 audited images, and 38–39 of 40 caliper-free images are clean. For the ovary, every detection is an on-screen overlay but only 31 of 40 are calipers (the others are dotted zoom arcs and focus markers), and the “marker-free” group is caliper-free but not overlay-free. The morphological hair detector of the pilot is poor (IoU 0.222), which is why the hair traps use the published masks.

Why label noise cannot create the result An image mislabelled as artifact-bearing (or artifact-free) weakens the constructed association between A and Y equally in both traps; it biases every effect toward zero. To create a crossover, errors would have to depend on location *and* on the label, which none of the detectors uses.

Table 3: Validity of the automatic masks that define artifact presence and location. Visual audits: docs/ARTIFACT_MASK_AUDIT.md and docs/THYROID_MARKER_AUDIT.md (visual audits during the analysis, not by a clinician).

Mask	Source	Check	Result
ISIC 2019 lesion masks (BCN, MSK)	U-Net trained on ISIC 2018 Task 1	held-out Dice (mean) / Dice vs HAM manual masks	0.886 / 0.895
Hair (DullRazor-type detector, E10)	morphological	IoU / precision / recall vs Mendeley masks	0.222 / 0.316 / 0.472
Capsule debris masks	patch-token probe (2,160 expert-masked frames)	held-out pixel accuracy / IoU	0.922 / 0.625
Thyroid (TN3K), detected (A1)	visual audit, 40 images	detection lies on a real caliper / measurement mark	38/40 (95 %); 2 border false positives; 1 dotted zoom box (overlay)
Thyroid, marker-free (A0)	visual audit, 40 images	no visible caliper	38–39/40 (95–97 %); depth-scale ticks present in all images (not label-related)
Ovary (MMOTU), detected (A1)	visual audit, 40 images	an on-screen overlay is present	40/40; caliper specifically 31/40 (others: dotted zoom arcs, focus markers, thumbnail inset)
Ovary, marker-free (A0)	visual audit, 40 images	no visible caliper	37/40 (92 %)
Ovary, marker-free (A0)	visual audit, 40 images	no overlay of any kind (arrows, boxes)	~29/40 (72 %); text labels common

5.2 The crossover with real artifacts

Table 4 and Figure 3. With DINOv2, masking is worth less for in-ROI artifacts in every cohort:

- **Dermoscopy hair.** Masking *harms* when the hair lies on the lesion (-0.049 [$-0.083, -0.018$]) and helps when it lies outside ($+0.098$ [$+0.060, +0.136$]); crossover $+0.147$ [$+0.111, +0.185$] (as

in Stage 1). After masking the Trap-A model still separates the correlated test (0.917) far better than the reversed one (0.460).

- **Thyroid calipers.** Masking helps in both traps but far more when the calipers lie outside the nodule (+0.348 [+0.307, +0.388] versus +0.109 [+0.085, +0.133]; crossover +0.239 [+0.196, +0.280]). In Trap A the masked model remains caliper-driven: correlated 0.908, reversed 0.398 (Table 7); in Trap B masking removes the gap (0.785 versus 0.752).
- **Ovarian calipers.** Crossover +0.170 [+0.087, +0.252] (Trap A +0.132 [+0.072, +0.196], Trap B +0.302 [+0.225, +0.378]).
- **Capsule debris.** The largest crossover (+0.377 [+0.327, +0.426]): masking recovers almost all of the reversed-test loss when the debris lies outside the lesion boxes (+0.461 [+0.415, +0.506]) and little when it lies on them (+0.084 [+0.055, +0.112]).

Other encoder families The crossover replicates with MedSigLIP (thyroid +0.195 [+0.153, +0.238], ovary +0.178 [+0.109, +0.248], capsule +0.335 [+0.286, +0.383]) and with the CNN ConvNeXt (thyroid +0.275 [+0.231, +0.320], capsule +0.320 [+0.268, +0.372]). With ConvNeXt, masking is useless inside the nodule (−0.004 [−0.029, +0.020]) and strongly helpful outside it (+0.271 [+0.228, +0.314]). The effect is not a property of vision transformers or of self-supervised training.

Harm or loss of benefit? Only for dermoscopy hair does masking actually harm in Trap A. For calipers and debris it still helps a little in Trap A — it removes clutter and overlays elsewhere on the screen — but a large residual shortcut remains. The universal real-data statement is therefore the *crossover*; harm appears where masking also removes disease context (dermoscopy, and in Stage 2 the controlled sweeps).

Table 4: Real-artifact traps: change in reversed-test AUROC from ROI masking, per trap, and the crossover (5 seeds $\times$ 5 grouped folds; crossed 95% CIs).

Cohort	Encoder	ERM reversed AUROC A / B	mask − ERM, Trap A (in ROI)	mask − ERM, Trap B (outside)	Crossover B − A
Dermoscopy hair	DINOv2	0.510 / 0.510	−0.049 [−0.083, −0.018]	+0.098 [+0.060, +0.136]	+0.147 [+0.111, +0.185]
Dermoscopy hair	DermLIP	0.665 / 0.735	−0.104 [−0.143, −0.067]	−0.090 [−0.135, −0.046]	+0.014 [−0.037, +0.063]
Thyroid calipers	DINOv2	0.289 / 0.405	+0.109 [+0.085, +0.133]	+0.348 [+0.307, +0.388]	+0.239 [+0.196, +0.280]
Thyroid calipers	MedSigLIP	0.289 / 0.481	+0.053 [+0.032, +0.073]	+0.248 [+0.200, +0.293]	+0.195 [+0.153, +0.238]
Ovarian calipers	DINOv2	0.413 / 0.423	+0.132 [+0.072, +0.196]	+0.302 [+0.225, +0.378]	+0.170 [+0.087, +0.252]
Capsule debris	DINOv2	0.581 / 0.429	+0.084 [+0.055, +0.112]	+0.461 [+0.415, +0.506]	+0.377 [+0.327, +0.426]
Capsule debris	MedSigLIP	0.570 / 0.464	+0.117 [+0.088, +0.147]	+0.452 [+0.407, +0.495]	+0.335 [+0.286, +0.383]
Thyroid calipers	ConvNeXt	0.357 / 0.429	−0.004 [−0.029, +0.020]	+0.271 [+0.228, +0.314]	+0.275 [+0.231, +0.320]
Ovarian calipers	MedSigLIP	0.426 / 0.551	+0.002 [−0.048, +0.051]	+0.180 [+0.120, +0.238]	+0.178 [+0.109, +0.248]
Capsule debris	ConvNeXt	0.528 / 0.431	+0.084 [+0.049, +0.118]	+0.403 [+0.358, +0.448]	+0.320 [+0.268, +0.372]

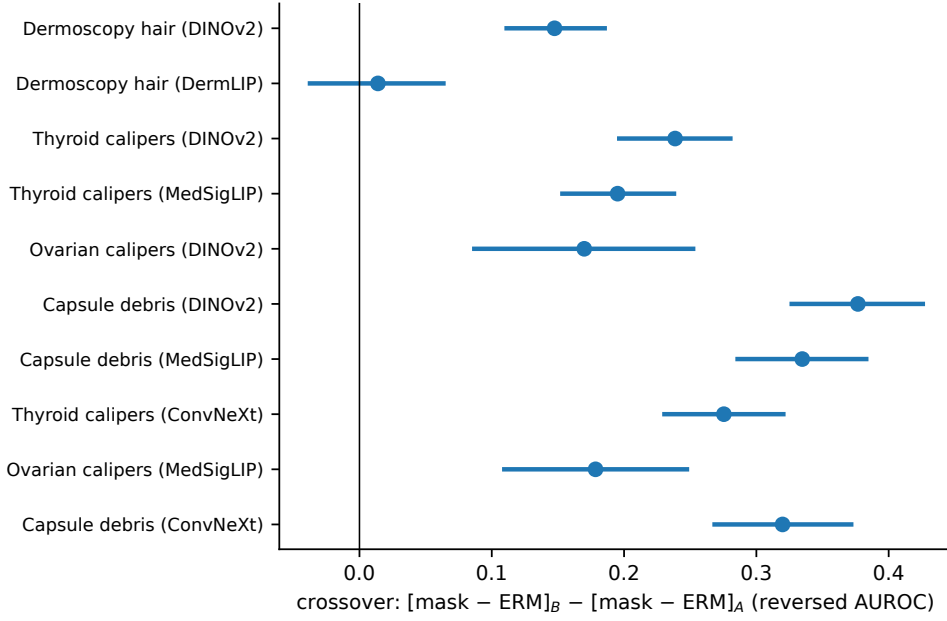


Figure 3: Crossover with real artifacts, per cohort and encoder (crossed 95% CIs).

5.3 The four location tests

Table 5. Each cohort’s crossover test was registered with the cohort; they are grouped here into one Holm family. All 4 of 4 are supported ($p \leq 0.0001$ before adjustment).

Table 5: The four pre-specified location tests (P1: crossover > 0), crossed bootstrap, two-sided bootstrap p (floored at 10^{-4}), Holm-adjusted over the four cohorts.

Cohort (DINOv2)	Crossover [95% CI]	p	p (Holm, 4)	Supported
ISIC hair (DINOv2)	+0.147 [+0.111, +0.185]	0.0001	0.0004	supported
Thyroid (DINOv2)	+0.239 [+0.196, +0.280]	0.0001	0.0004	supported
Capsule (DINOv2)	+0.377 [+0.327, +0.426]	0.0001	0.0004	supported
Ovary (DINOv2)	+0.170 [+0.087, +0.252]	0.0001	0.0004	supported

5.4 Every arm, and the environments

Tables 6 and 7. Three observations matter.

- Label-based methods fix both traps.** Group balancing and DFR, which need image-level artifact labels, help in both traps of every cohort (e.g. thyroid DFR +0.463 [+0.436, +0.492] in Trap A), and their benefit does not depend on location. This replicates the pilot’s third claim on three new cohorts.
- Oracle inpainting fails for capsule debris.** Inpainting the debris pixels *lowers* reversed AUROC (−0.172 [−0.202, −0.142] in Trap A). Erosions carry yellow-white fibrin that resembles debris; inpainting what the probe calls debris removes disease evidence. Removing artifact pixels is only safe when the artifact does not resemble the disease.
- Two arms “help” by flipping the shortcut.** Unpaired LEACE and prevalence calibration show the largest reversed-test gains (+0.515 [+0.491, +0.538], +0.602 [+0.580, +0.625] for thyroid), but both remove or invert the artifact–label association using the artifact label of the *training* set (and prevalence calibration also at test time), so their correlated AUROC falls far below their reversed one — the same failure as in Stage 1. They are reported, not recommended. Paired LEACE, the sound variant, recovers little (thyroid +0.052 [+0.045, +0.059]).

Table 6: Every baseline arm in the real-artifact traps (DINOv2 @518, 5 seeds $\times$ 5 folds): change in reversed-test AUROC versus ERM, crossed 95% CIs. Inpainting needs the artifact mask; balancing, DFR, unpaired LEACE and prevalence calibration need image-level artifact labels (prevalence calibration also at test time).

Cohort	Arm – ERM	Trap A (in ROI)	Trap B (outside)
Dermoscopy hair	masking	−0.049 [−0.083, −0.018]	+0.098 [+0.060, +0.136]
	oracle inpainting	+0.098 [+0.084, +0.112]	+0.100 [+0.083, +0.117]
	group-balanced	+0.213 [+0.183, +0.243]	+0.204 [+0.166, +0.245]
	DFR	+0.196 [+0.157, +0.235]	+0.196 [+0.147, +0.251]
	paired LEACE	+0.074 [+0.062, +0.086]	+0.078 [+0.059, +0.097]
	unpaired LEACE	+0.320 [+0.280, +0.361]	+0.295 [+0.253, +0.336]
	prevalence calibration	+0.415 [+0.377, +0.455]	+0.429 [+0.391, +0.466]
Thyroid calipers	masking	+0.109 [+0.085, +0.133]	+0.348 [+0.307, +0.388]
	oracle inpainting	+0.102 [+0.092, +0.112]	+0.039 [+0.025, +0.053]
	group-balanced	+0.388 [+0.360, +0.418]	+0.254 [+0.218, +0.292]
	DFR	+0.463 [+0.436, +0.492]	+0.312 [+0.262, +0.362]
	paired LEACE	+0.052 [+0.045, +0.059]	+0.023 [+0.014, +0.032]
	unpaired LEACE	+0.515 [+0.491, +0.538]	+0.338 [+0.302, +0.375]
	prevalence calibration	+0.602 [+0.580, +0.625]	+0.501 [+0.463, +0.538]
Ovarian calipers	masking	+0.132 [+0.072, +0.196]	+0.302 [+0.225, +0.378]
	oracle inpainting	+0.160 [+0.135, +0.186]	+0.046 [+0.032, +0.061]
	group-balanced	+0.290 [+0.248, +0.332]	+0.256 [+0.190, +0.320]
	DFR	+0.310 [+0.263, +0.357]	+0.278 [+0.209, +0.349]
	paired LEACE	+0.142 [+0.119, +0.166]	+0.031 [+0.020, +0.043]
	unpaired LEACE	+0.369 [+0.320, +0.418]	+0.305 [+0.230, +0.375]
	prevalence calibration	+0.514 [+0.468, +0.561]	+0.507 [+0.440, +0.571]
Capsule debris	masking	+0.084 [+0.055, +0.112]	+0.461 [+0.415, +0.506]
	oracle inpainting	−0.172 [−0.202, −0.142]	−0.115 [−0.144, −0.087]
	group-balanced	+0.179 [+0.150, +0.205]	+0.244 [+0.216, +0.272]
	DFR	+0.234 [+0.199, +0.273]	+0.287 [+0.239, +0.335]
	paired LEACE	+0.012 [+0.004, +0.020]	+0.010 [+0.001, +0.024]
	unpaired LEACE	+0.282 [+0.247, +0.316]	+0.416 [+0.374, +0.457]
	prevalence calibration	+0.347 [+0.316, +0.376]	+0.465 [+0.430, +0.498]

Table 7: AUROC of ERM, masking, balancing and DFR in the three test environments (DINOv2, mean over 5 seeds). A large correlated–reversed gap is reliance on the artifact.

Cohort	Trap	Arm	clean	correlated	reversed
Dermoscopy hair	Trap A	erm	0.734	0.920	0.510
	Trap A	mask	0.712	0.917	0.460
	Trap A	balanced	0.785	0.850	0.722
	Trap A	dfr	0.715	0.740	0.705
	Trap B	erm	0.716	0.894	0.510
	Trap B	mask	0.701	0.795	0.608
	Trap B	balanced	0.752	0.800	0.714
	Trap B	dfr	0.683	0.669	0.706
Thyroid calipers	Trap A	erm	0.608	0.879	0.289
	Trap A	mask	0.689	0.908	0.398
	Trap A	balanced	0.717	0.757	0.677
	Trap A	dfr	0.708	0.669	0.752
	Trap B	erm	0.600	0.789	0.405
	Trap B	mask	0.767	0.785	0.752
	Trap B	balanced	0.657	0.660	0.659
	Trap B	dfr	0.653	0.582	0.716
Ovarian calipers	Trap A	erm	0.635	0.826	0.413
	Trap A	mask	0.690	0.828	0.545
	Trap A	balanced	0.710	0.725	0.703
	Trap A	dfr	0.708	0.689	0.723
	Trap B	erm	0.636	0.815	0.423
	Trap B	mask	0.733	0.752	0.725
	Trap B	balanced	0.694	0.710	0.679
	Trap B	dfr	0.671	0.632	0.701
Capsule debris	Trap A	erm	0.796	0.941	0.581
	Trap A	mask	0.860	0.972	0.665
	Trap A	balanced	0.847	0.916	0.760
	Trap A	dfr	0.842	0.866	0.816
	Trap B	erm	0.714	0.922	0.429
	Trap B	mask	0.932	0.962	0.890
	Trap B	balanced	0.784	0.879	0.673

Cohort	Trap	Arm	clean	correlated	reversed
	Trap B	dfr	0.762	0.819	0.716

5.5 What carries the hair shortcut?

Hair presence and location go with patient and body site (Table 8): in HAM, hair-free lesions are more often from women and from the lower extremity; in BCN, in-lesion hair goes with older patients and the head and neck. Two diagnostics show that the hair shortcut is largely carried by these correlates rather than by hair pixels:

- **Pixel share.** Removing the hair pixels at test time only (oracle inpainting of the ERM model’s test images) closes 47% (Trap A) and 44% (Trap B) of ERM’s clean–reversed gap: more than half of what ERM exploits is not in the hair pixels.
- **Oracle inpainting** gains about +0.098 [+0.084, +0.112] in Trap A, half of what balancing gains (+0.213 [+0.183, +0.243]), which removes the association for hair and everything correlated with it.

Two archived sensitivity designs point the same way (**point estimates, not re-estimated**): an author-independent reconstruction of the traps (strictly hair-free comparison group, artifact-free clean test) gives a crossover of +0.166 but *no* Trap-A harm (+0.012); traps matched on site, sex and age band give +0.169 (Trap A +0.030). The crossover is robust to these choices; the in-lesion harm depends on including lightly haired images and on the strict hair-free group. Hair is therefore the *correlate-carried* regime: masking cannot remove a shortcut that is carried by who the patient is and where the lesion is.

Table 8: Patient and site metadata of the hair groups (ISIC 2019): hair presence and location go with age, sex and body site.

Source	Group	Mean age	Share female	Share head/neck	Share lower extremity
HAM	hair-free	57.9	0.57	0.10	0.42
	Trap A (on lesion)	52.1	0.42	0.12	0.16
	Trap B (beside)	49.9	0.45	0.14	0.26
BCN	hair-free	61.3	0.59	0.18	0.32
	Trap A (on lesion)	58.9	0.45	0.33	0.13
	Trap B (beside)	51.8	0.45	0.22	0.18
MSK	hair-free	51.2	0.41	0.07	0.14
	Trap A (on lesion)	54.5	0.33	0.09	0.08
	Trap B (beside)	46.9	0.45	0.09	0.11

Table 9: Dermoscopy hair, Trap A, by image source (DINOv2).

Source	mask – ERM, Trap A
HAM	−0.011 [−0.064, +0.043]
BCN	−0.067 [−0.110, −0.025]

Hair area relative to the lesion In Trap A the in-lesion hair covers a median 0.033 of the lesion’s area — similar to the synthetic ruler of Stages 1–2. Masking harms in every tertile of that ratio (Table 10); unlike the synthetic sweep, no gradient is visible — consistent with a shortcut that is not mainly in the hair pixels.

Table 10: Dermoscopy hair, Trap A, by the ratio of in-lesion hair area to lesion area.

Hair/lesion area tertile	Median ratio	mask – ERM, Trap A
tertile 1	0.006	−0.057 [−0.095, −0.022]
tertile 2	0.033	−0.046 [−0.082, −0.013]
tertile 3	0.113	−0.037 [−0.071, −0.005]

5.6 The exception among encoders: DermLIP

With DermLIP masking harms at *both* hair locations (Trap A −0.104 [−0.143, −0.067], Trap B −0.090 [−0.135, −0.046]) and the crossover vanishes (+0.014 [−0.037, +0.063]). DermLIP was trained on whole dermatology images and loses disease signal when the background is removed, whatever the hair does (Stage 1: clean AUROC falls with masking in both traps).

5.7 Chest radiography: the boundary

Device traps Table 11. With the registered primary encoder RAD-DINO, lung masking helped for both device types in all four findings (X1 in 4 of 4); neither the in-lung penalty (X2) nor the crossover (X3) was supported (crossovers −0.025 to +0.046; in the archived per-seed analysis one finding, atelectasis, had passed X3). With the registered replication encoder MedSigLIP, fitted for the first time when these runs were regenerated, the crossover is positive for all four findings (+0.095 to +0.328, Holm), and for effusion masking harms inside the lungs (−0.084). The two encoders differ in what they use: MedSigLIP relies heavily on the out-of-lung tube, which masking removes (Trap B gains of 0.23–0.31), whereas RAD-DINO barely uses extra-pulmonary tubes (the synthetic-tube sweep of Stage 2). A confound noted before the follow-up was fitted: the catheter-free group carries many more out-of-lung tubes than the catheter group (endotracheal tubes 77% versus 39%, nasogastric tubes 83% versus 42%), because CLiP’s catheter-free images were selected for other devices; masking removes those tubes, which inflates masking’s gain in Trap A and so works against a positive crossover. A RAD-DINO follow-up matched within label on view and on the presence of the other devices (Infiltration) still shows no crossover (Trap A +0.089, Trap B +0.042, crossover −0.047; archived point estimates, **not re-estimated**). The registered DINOv2 replication was not run.

Table 11: Chest radiographs with real devices (NIH ChestX-ray14 linked to RANZCR-CLiP, AP films): Trap A = central venous catheter inside the lungs, Trap B = endotracheal tube outside them; reversed-test AUROC, crossed 95% CIs, Holm over the four findings per encoder. Registered claims: X1 masking helps in Trap B, X2 masking harms in Trap A, X3 crossover > 0. RAD-DINO is the registered primary encoder; the registered MedSigLIP replication was fitted for the first time when these runs were regenerated.

Encoder	Finding	mask – ERM, Trap A (in lungs)	mask – ERM, Trap B (outside)	Crossover	<i>p</i> (Holm, 4)	Supported
RAD-DINO	Atelectasis	+0.059 [+0.019, +0.099]	+0.105 [+0.092, +0.117]	+0.046 [+0.011, +0.082]	0.0544	X1
	Consolidation	+0.074 [+0.025, +0.125]	+0.070 [+0.047, +0.092]	−0.004 [−0.065, +0.058]	0.9112	X1
	Effusion	+0.078 [+0.032, +0.125]	+0.100 [+0.089, +0.111]	+0.022 [−0.030, +0.074]	0.8364	X1
	Infiltration	+0.087 [+0.060, +0.116]	+0.063 [+0.055, +0.071]	−0.025 [−0.055, +0.006]	0.3444	X1
MedSigLIP	Atelectasis	+0.183 [+0.146, +0.220]	+0.278 [+0.261, +0.296]	+0.095 [+0.053, +0.137]	0.0004	X1 X3
	Consolidation	+0.087 [+0.031, +0.142]	+0.226 [+0.206, +0.246]	+0.139 [+0.081, +0.197]	0.0004	X1 X3
	Effusion	−0.084 [−0.125, −0.043]	+0.244 [+0.230, +0.259]	+0.328 [+0.287, +0.369]	0.0004	X1 X2 X3
	Infiltration	+0.174 [+0.128, +0.219]	+0.309 [+0.293, +0.325]	+0.135 [+0.084, +0.185]	0.0004	X1 X3

Chest drains Table 12. The drain is an extreme shortcut: ERM’s reversed AUROC is 0.194 (RAD-DINO) and 0.284 (DINOv2), against clean 0.888 and 0.822. Lung masking barely helps (0.232, 0.289; gain over ERM +0.039 [+0.019, +0.058] and +0.005 [−0.013, +0.023]), because the drain lies in the lungs. Label-based DFR is best (0.666, 0.537): a drain comes with a treated, re-expanded lung, so the shortcut is carried by the treatment’s consequences, not only by the tube

— a second correlate-carried case.

Table 12: Real chest drains (NIH ChestX-ray14 pneumothorax, 29,687 images, patient-grouped folds; drains lie inside the lungs, so this is an in-ROI trap only). AUROC, mean over 5 seeds of 5 patient-grouped folds.

Encoder	Arm	clean	correlated	reversed
RAD-DINO	ERM	0.888	0.946	0.194
	lung masking	0.873	0.947	0.232
	group-balanced	0.858	0.880	0.540
	DFR	0.802	0.758	0.666
	masking + DFR	0.802	0.779	0.635
DINOv2	ERM	0.822	0.920	0.284
	lung masking	0.833	0.931	0.289
	group-balanced	0.786	0.846	0.402
	DFR	0.771	0.806	0.537
	masking + DFR	0.788	0.828	0.561

Reading Together with the synthetic-tube sweep of Stage 2 (RAD-DINO ignores an out-of-lung tube), chest radiography is a boundary: in-lung devices are confounded with other devices or carried by correlates, and the radiograph encoder does not use extra-pulmonary tubes at all. A general medical encoder that does use them (MedSigLIP) shows the crossover, which is consistent with the law, but one encoder with a confounded contrast is not enough to count chest radiography as a fifth confirmation; it is reported as a boundary case.

5.8 Datasets considered and not used

Table 13: Cohorts evaluated for a real-artifact trap and rejected, with the reason (decided before any model was fitted on them).

Dataset	Artifact	Reason
Breast ultrasound (BUSI, BUS-BRA), attempt 1	calipers	the thyroid detector misses GE-style “x” calipers; a two-detector design left the artifact-free group only $\approx 70\%$ pure on audit
Breast ultrasound, attempt 2	calipers	a patch-token caliper probe (AUROC 0.98 on thyroid/ovary) found only 19 of 2,522 images free of on-screen markings: no artifact-free group exists
CANDID-PTX	chest drains	access not granted
CXR cardiomegaly lines	lines	too few labelled images
TCGA / GrandQC pathology	pen marks	pen masks unreliable
ISIC ink	ink marks	ink on the lesion in only 70 images
EAD endoscopy	specular reflections	no disease labels
BKAI-IGH NeoPolyp	specular highlights	no specular-free images (0), 41 in-polyp
colonoscopy		
DINOv3 encoder	—	weights gated, access refused

5.9 Sensitivity to the interval estimator

On the traps of this stage (Table 14) the crossed intervals are wider by a median factor of 1.62; 6 of 159 contrasts lose significance and 3 gain it (Table 15); none of them is a crossover or a masking effect on the reversed test.

Table 14: Real-artifact traps of the new cohorts under the per-seed and the crossed estimator (same predictions; baseline arms only).

Result group	Intervals	Excluded with the estimator	0 only per-seed	Excluded with the crossed estimator	Median width ratio (crossed / per-seed)
capsule/convnext384_universal	17	0		0	1.38
capsule/dino518_main	25	0		1	1.59
capsule/medsiglip448_main	25	0		0	1.69
ovary/dino518_main	25	2		0	1.67
thyroid/convnext384_universal	17	3		0	1.68
thyroid/dino518_main	25	1		1	1.62
thyroid/medsiglip448_main	25	0		1	1.57
all	159	6		3	1.62

Table 15: Contrasts of the real-artifact traps whose verdict depends on the estimator.

Result	Contrast	Crossed estimate [95% CI]	Verdict vs per-seed estimator
thyroid/dino518_main	trapB / dfr / clean / dfr / erm	+0.053 [+0.002, +0.105]	now excludes 0
thyroid/dino518_main	trapB / leace_paired / clean / leace_paired / erm	+0.008 [−0.001, +0.018]	no longer excludes 0
thyroid/medsiglip448_main	trapA / inpaint / clean / inpaint / erm	+0.015 [+0.003, +0.025]	now excludes 0
thyroid/convnext384_universal	trapA / mask / clean / mask / erm	+0.021 [−0.000, +0.041]	no longer excludes 0
thyroid/convnext384_universal	trapB / balanced / clean / balanced / erm	+0.039 [−0.002, +0.080]	no longer excludes 0
thyroid/convnext384_universal	trapB / dfr / clean / dfr / erm	+0.052 [−0.001, +0.103]	no longer excludes 0
ovary/dino518_main	trapB / dfr / clean / dfr / erm	+0.035 [−0.018, +0.089]	no longer excludes 0
ovary/dino518_main	trapB / leace_paired / clean / leace_paired / erm	+0.008 [−0.000, +0.015]	no longer excludes 0
capsule/dino518_main	trapB / leace_paired / test_rev / leace_paired / erm	+0.010 [+0.001, +0.024]	now excludes 0

6 What failed or changed

- **In-ROI masking harms only for dermoscopy hair**; for calipers and debris it helps less. The registered “harm” claims (T2, O2, C2) were not supported for thyroid, ovary and capsule.
- **DermLIP has no hair crossover**; “the crossover holds for every encoder” is not claimed.
- **Chest-radiograph device traps**: with RAD-DINO the registered in-ROI penalty was not supported (crossover in 0 of 4 findings after regeneration, 1 of 4 in the archived analysis); the device-matched follow-up, which was not regenerated, does not change this. With the MedSigLIP replication it is supported (crossover in 4 of 4, in-lung harm in 1 of 4).
- **Oracle inpainting fails on capsule debris**, where the artifact resembles the disease.
- **The hair shortcut is mostly correlate-carried**, and the in-lesion harm is sensitive to how the hair-free group is defined.
- **Breast ultrasound was attempted twice and abandoned** (no artifact-free group).
- **The artifact masks were audited visually by a non-clinician**; no clinician has checked them yet.

7 Anticipated questions

Why not move the real artifact, as in the sweeps? A real artifact’s position is part of the image. The two-trap design compares images whose artifact happens to lie inside with images whose artifact lies outside; both are completed with the same artifact-free images, so the only designed

difference is location. That the two groups of images may differ in other ways is a real limitation of this design.

Why $r \geq 0.5$ and $r < 0.1$, and what about the images in between? The thresholds come from the pilot and leave a gap so that the two traps are clearly separated; the in-between images are used by no trap.

Why does masking still help in Trap A for calipers and debris? Ultrasound and capsule frames carry other on-screen marks and clutter outside the ROI, and removing them helps. What masking cannot do is remove the caliper on the nodule; the masked Trap-A model’s correlated–reversed gap stays large.

Why is thyroid benign-heavy in Trap A? Calipers are drawn on nodules that are measured, and in this cohort measured nodules are more often benign; the environments fix the association at 0.9/0.1 whatever the natural rate.

Why are balancing and DFR not simply the answer? They need image-level artifact labels for every training image, which clinical datasets rarely have; masking needs only an ROI. They are the reference that shows what is achievable when labels exist.

Why keep DermLIP if it contradicts the law? It is a pre-registered replication, and its failure is informative: an encoder that needs the surroundings of a lesion loses disease signal with any mask, so the comparison between locations is dominated by that loss.

Why is chest radiography not counted as a confirmation? In the device traps the in-lung catheter is confounded with other devices, and in the drain trap the shortcut is carried by the treated lung. Neither isolates location; counting them either way would be unjustified. The MedSigLIP replication points towards the law, and the confound works against it, but the primary radiograph encoder shows no crossover, so the evidence is split by encoder.

Is Holm over four tests enough? The four p values are at the bootstrap floor (10^{-4}); any multiplicity correction over any plausible family leaves them significant.

How reliable are the capsule masks at IoU 0.625? Pixel IoU is modest, but the traps use thresholds on the *fraction* of the frame and of the lesion box covered; small boundary errors move few frames across 3% or 10%. Errors dilute effects rather than create them (Section on mask validity).

8 Conclusion

Supported. With real artifacts, masking is worth less when the artifact lies inside the ROI: the crossover is positive in all four cohorts with DINOv2 (Holm over the four location tests), with MedSigLIP in three cohorts and with a CNN in two. Label-based methods fix both traps. **Qualified.** Only for dermoscopy hair does masking actually harm in the in-ROI trap, and that harm depends on how the hair-free group is defined; hair is a correlate-carried shortcut. **Not supported.** A crossover with DermLIP; an in-ROI penalty for real chest-radiograph devices with RAD-DINO (supported with MedSigLIP); pixel removal for debris that resembles the disease.

9 Questions for the supervisor

1. Is a visual audit of the automatic artifact masks by a non-clinician acceptable for the thesis, or should a clinician check a sample before the real-artifact results are written up?
2. Chest radiography now has crossed intervals and a positive MedSigLIP replication (crossover in all four findings) beside a null primary encoder. Should it stay a boundary case, become a qualified fifth cohort, or move to an appendix?
3. How should the thesis present the correlate-carried nature of the hair shortcut: as a limitation of the hair cohort, or as a finding about real shortcuts in its own right?
4. Is it acceptable to count calipers and debris as confirmations when masking still helps (less) inside the ROI, i.e. to define the law by the crossover rather than by harm?

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